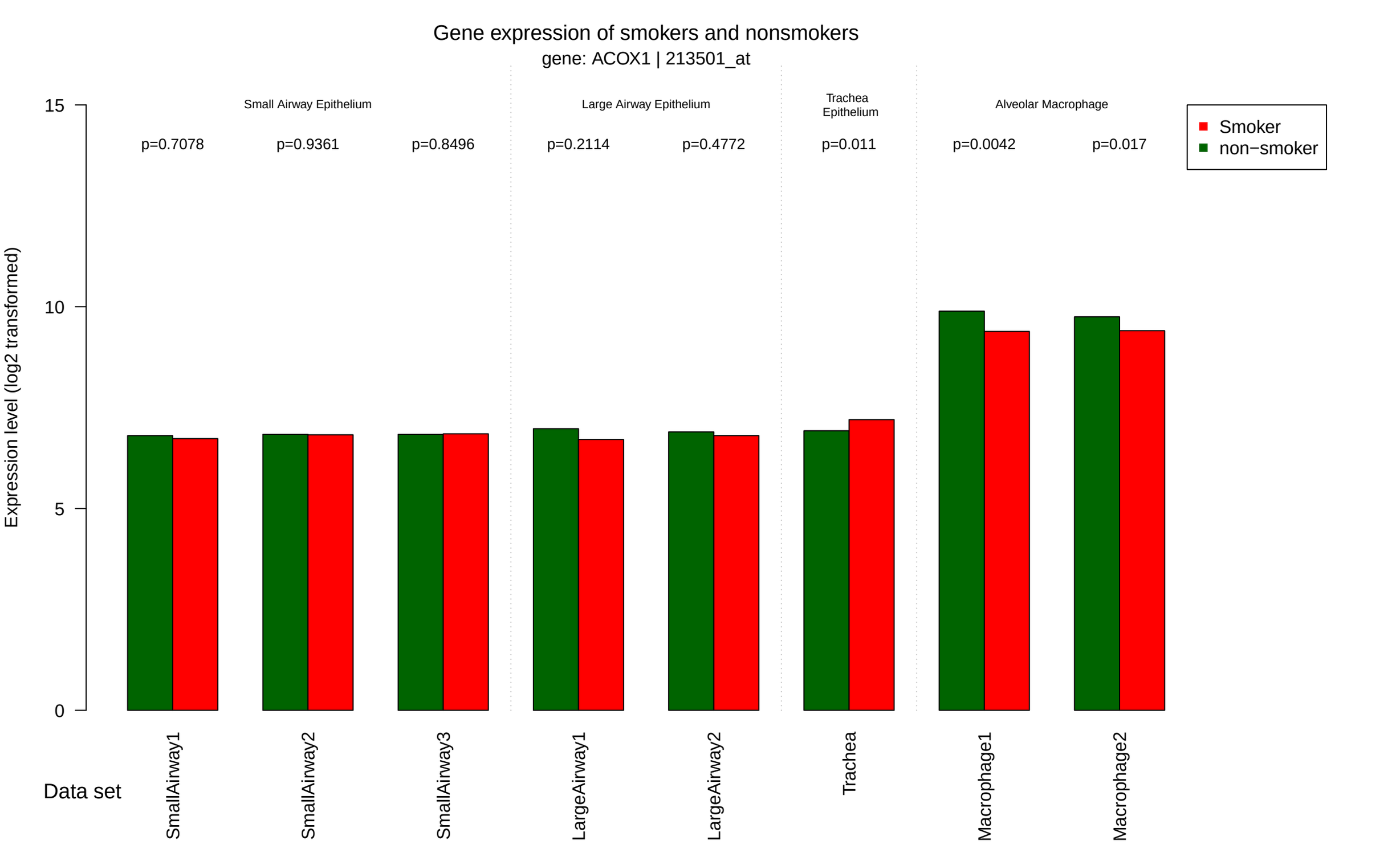
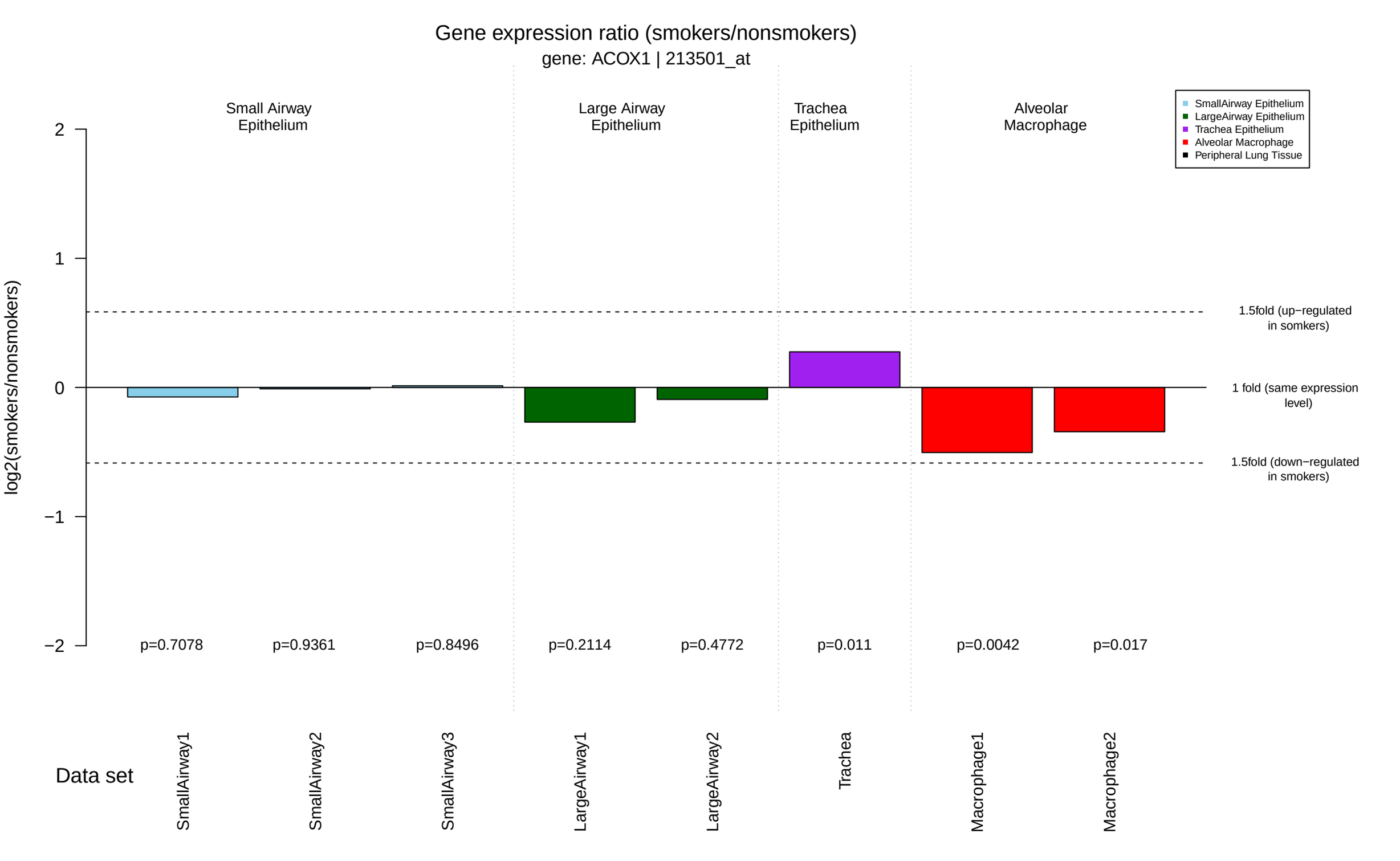
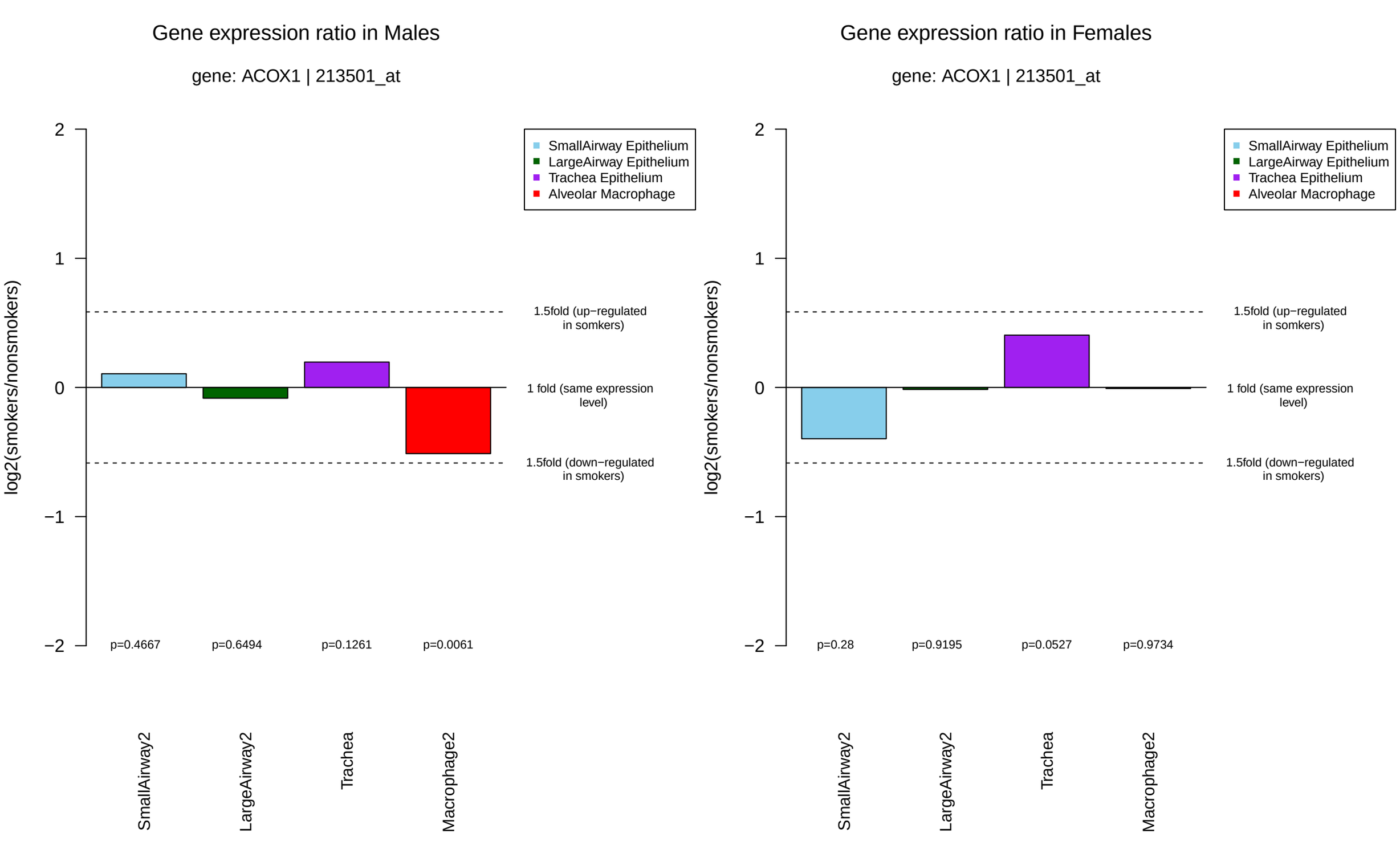
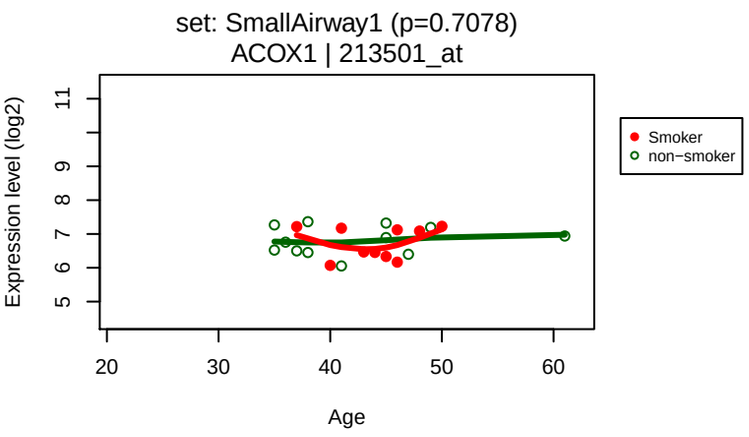
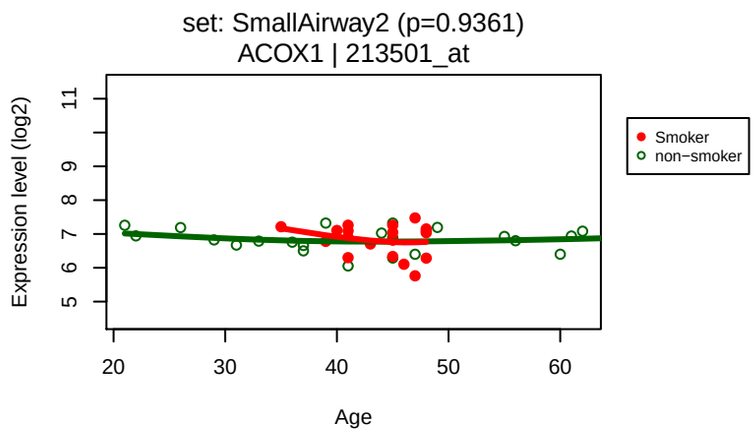
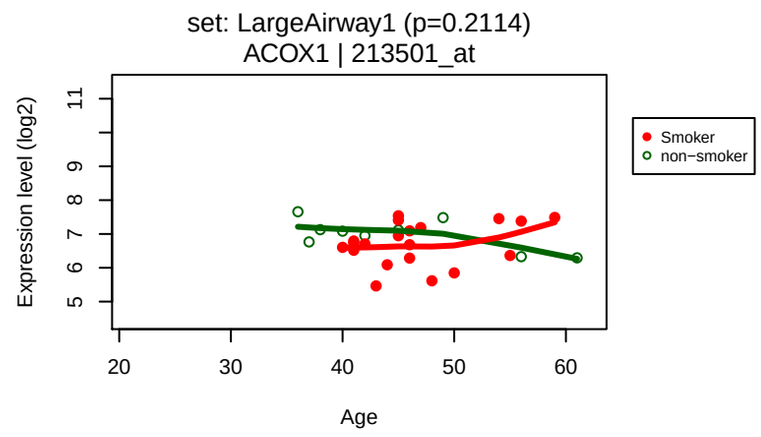

Gene expression of smokers and nonsmokers
gene: ACOX1 | 213501_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7078
p=0.9361
p=0.8496
p=0.2114
p=0.4772
p=0.011
p=0.0042
p=0.017
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

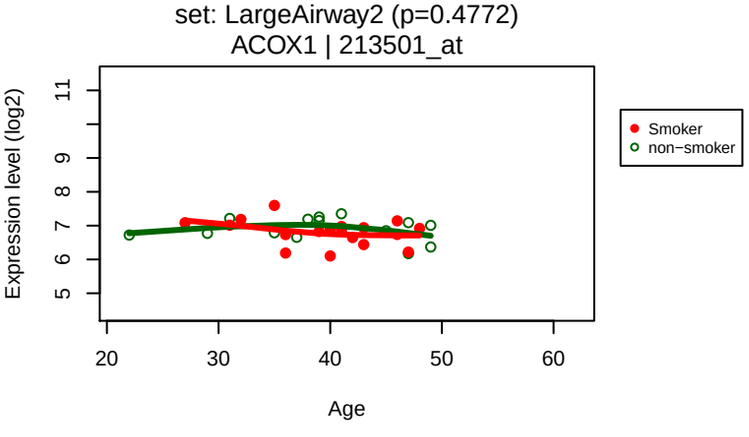

Gene expression ratio (smokers/nonsmokers)
gene: ACOX1 | 213501_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7078
p=0.9361
p=0.8496
p=0.2114
p=0.4772
p=0.011
p=0.0042
p=0.017
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

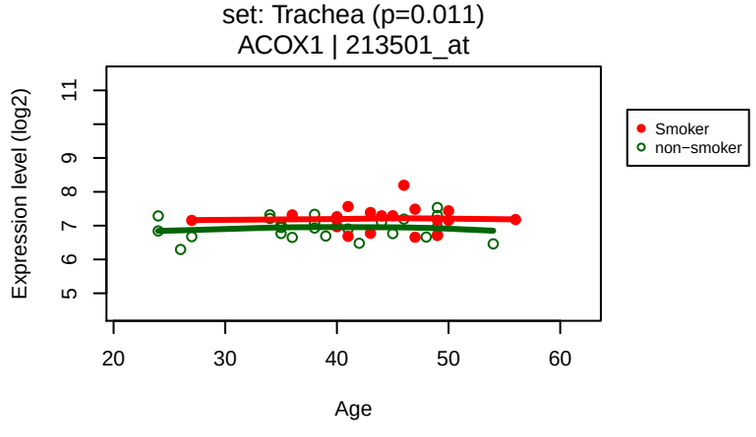

Gene expression ratio in Males
gene: ACOX1 | 213501_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4667
p=0.6494
p=0.1261
p=0.0061
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACOX1 | 213501_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.28
p=0.9195
p=0.0527
p=0.9734
SmallAirway2
LargeAirway2
Trachea
Macrophage2

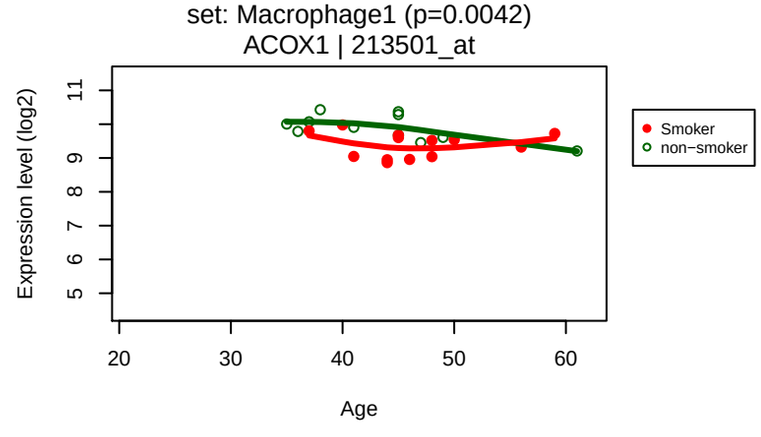

set: SmallAirway1 (p=0.7078)
ACOX1 | 213501_at
Expression level (log2)
Age
Smoker
non−smoker

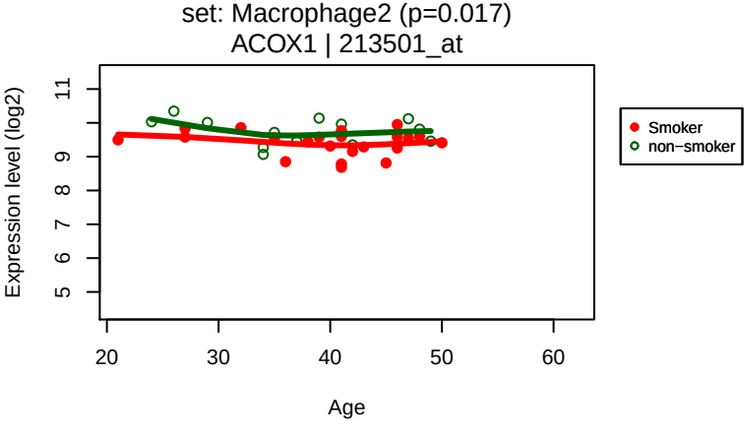

set: SmallAirway2 (p=0.9361)
ACOX1 | 213501_at
Expression level (log2)
Age
Smoker
non−smoker

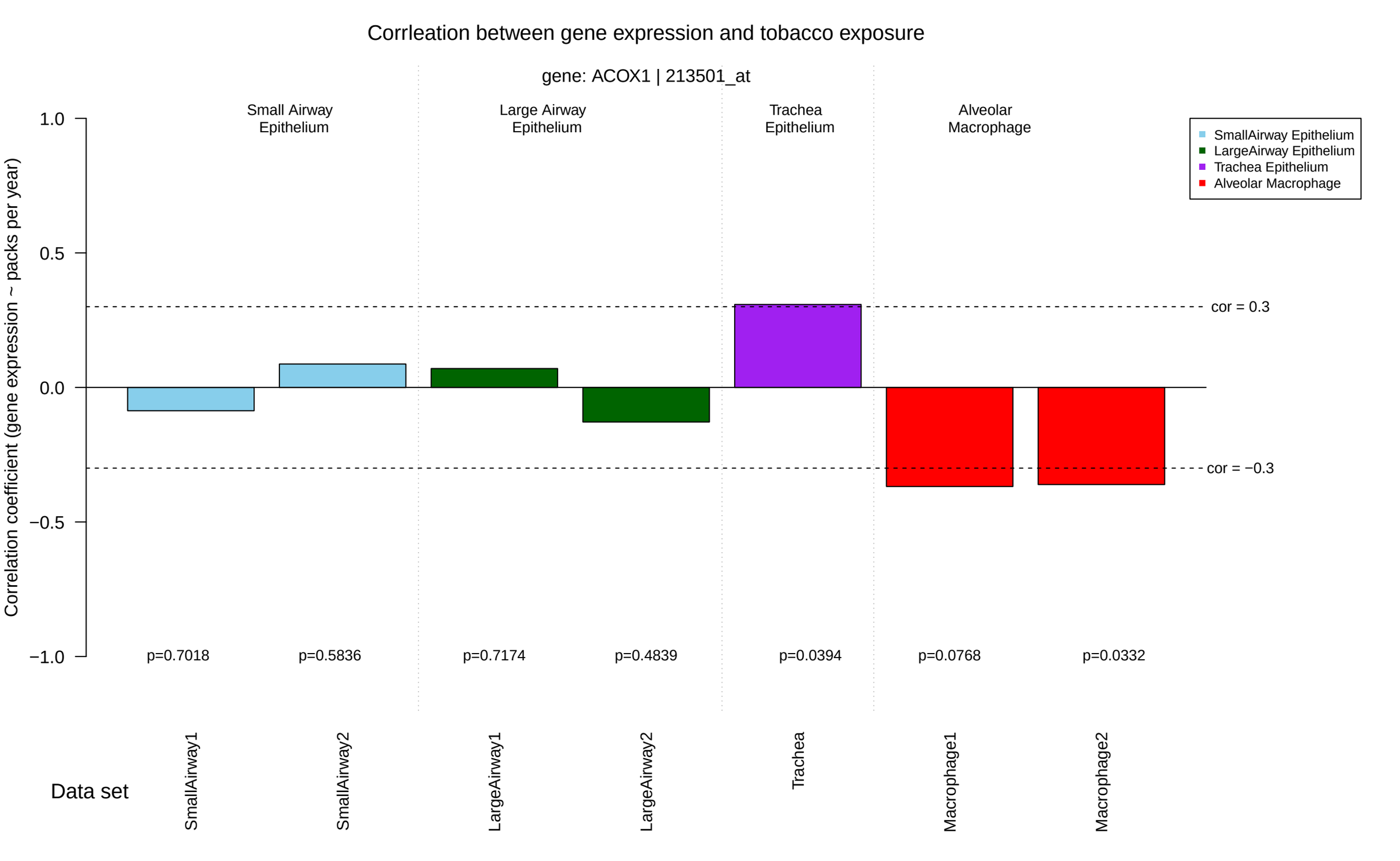

set: LargeAirway1 (p=0.2114)
ACOX1 | 213501_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4772)
ACOX1 | 213501_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.011)
ACOX1 | 213501_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0042)
ACOX1 | 213501_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.017)
ACOX1 | 213501_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACOX1 | 213501_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7018
p=0.5836
p=0.7174
p=0.4839
p=0.0394
p=0.0768
p=0.0332
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2